NEWTON SCHOOL

PRE-CALCULUS | TEST PREP

Test Prep Worksheet — Induction · Recursion · Series · Bernoulli

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Standard: *CCSS.MATH.CONTENT.HSF-IF.A.3 (Recursive sequences) · HSA-SSE.B.4 (Finite geometric series) · AP Calculus BC Topic 10 (Series · Convergence).*

Test Prep — Mr. Jones

Work each problem in full on the lines below. Show every induction step (base · hypothesis · inductive step), every recurrence (state base cases), and every series sum (formula · substitution · simplification). Box your final answer.

Topic 1 — Mathematical Induction

Directions: *Use the standard induction format — Base case · Inductive Hypothesis · Inductive Step · Conclusion. For #3, use strong induction.*

1. Prove by induction that $1^3 + 2^3 + 3^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$ for all integers $n \geq 1$.

2. Prove by induction that $13^n - 5^n$ is divisible by 8 for all integers $n \geq 1$.

3. Prove by **strong induction** that every integer $n \geq 2$ can be written as a product of (one or more) prime numbers.

Topic 2 — Run-of-Heads · Recursion

Directions: *For each recurrence, clearly state the base cases, the recursive relation, and the conditions on n. Reduce all probabilities to lowest terms.*

4. Let $p(n)$ be the probability that in n fair coin flips there is **no run of 3 heads in a row**. Set up the recurrence for $p(n)$ by splitting on how the sequence begins (T, HT, or HHT). State the values $p(1)$, $p(2)$, and $p(3)$.

5. Using the recurrence from #4, compute $p(5)$ as a fraction in lowest terms.

6. Let $a(n)$ be the number of binary strings of length n with **no two consecutive 1's**. Write the recurrence (with base cases) and compute $a(6)$.

Topic 3 — Series · Σ · Domain of Convergence

Directions: *Use the appropriate arithmetic / finite-geometric / infinite-geometric formula. For #8, state the inequality on x that guarantees $|r| < 1$ and give the closed-form sum on that domain.*

7. Find the sum $\sum_{k=1}^{20}(3k-2)$. (Arithmetic.)

8. Determine the **domain of convergence** (the set of x for which the series converges) and give the sum in closed form for

$$5 + 5\left(\frac{x+1}{2}\right) + 5\left(\frac{x+1}{2}\right)^2 + 5\left(\frac{x+1}{2}\right)^3 + \cdots$$

9. A geometric series has first term $a = 6$ and common ratio $r = \frac{2}{3}$. Find S_∞, and find the smallest n such that S_n is within 0.01 of S_∞.

Topic 4 — Bernoulli's Madness

Directions: *Use sums of powers, the hockey stick identity, telescoping, and the closed-form trick for the linear recurrence* $T_n = a\,T_{n-1} + b$. *Express each answer in closed form, then evaluate.*

10. Use the **hockey stick identity** to evaluate $\sum_{k=1}^{N} k(k+1)(k+2)$ in closed form.

11. Use **telescoping** to evaluate $\sum_{k=1}^{N} \frac{1}{k(k+1)}$.

12. Solve the linear recurrence $T_n = 2\,T_{n-1} + 3$ with $T_0 = 1$. Find a closed form for T_n and compute T_5.

Answer key: see paired lecture note (*JustinKoo-2026-05-13-...-LectureNote.pdf*).